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Unit 1 : Session 8 : SLO 2 | Complexity Time and Space Trade-off

I. Fill in the Blanks

1. Time is the amount of time an algorithm takes.
2. Space is the amount of memory an algorithm uses.
3. time and space is the trade-off balances the two.
4. time is more memory (e.g. caching) can reduce time complexity.
5. slowly (longer) is using less space may cost more time.
6. memoization (caching) is storing precomputed values to save time.
7. operations (elementary steps) is how time complexity is measured.
8. input is space complexity includes auxiliary space plus input space.
9. space (stack) is recursion uses call-stack memory, unlike iteration.
10. constant is $O(1)$ space uses a fixed amount of memory.
11. trade-off is sorting algorithms show clear time/space trade-offs.
12. iteratively is iterative binary search is $O(\log n)$ time, $O(1)$ space.
13. computational (algorithmic) is both are forms of computational complexity.
14. computations (subproblems) is DP stores results to avoid recomputation.
15. balance (trade-off) is optimization seeks the best time-space balance.